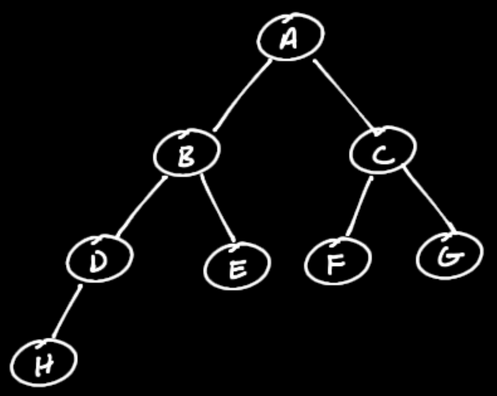
① Tree Terminology



Node
Edges
Root
Interior node
Leaf
Parent
Child
Sibling
Descendents
Ancestors
Path
Height
Subtree
Level

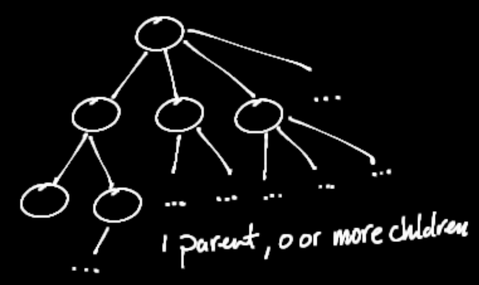
Stacks, queues, and list are linear and position oriented

Sorted list are linear and value oriented

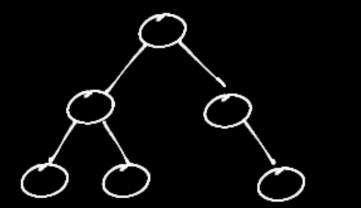
trees are non-linear and value oriented (for most cases)

→

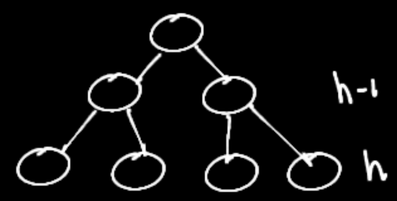
② General Tree



Binary Tree

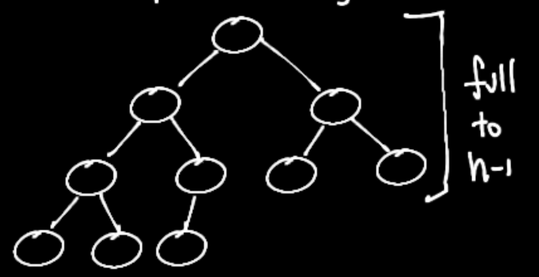


Full Binary Tree



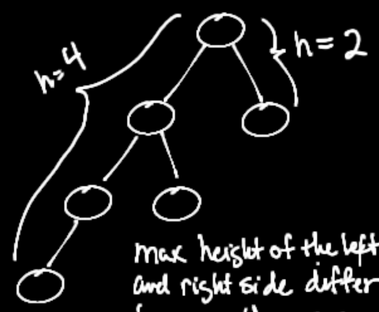
no "missing" nodes
all leafs at level h
all non-leafs have 2 children

Complete Binary Tree



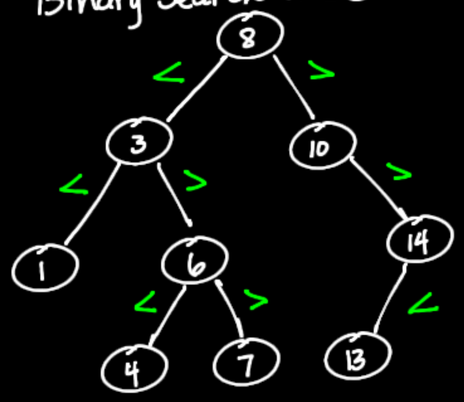
level h filled "left to right"

③ Non-balanced tree

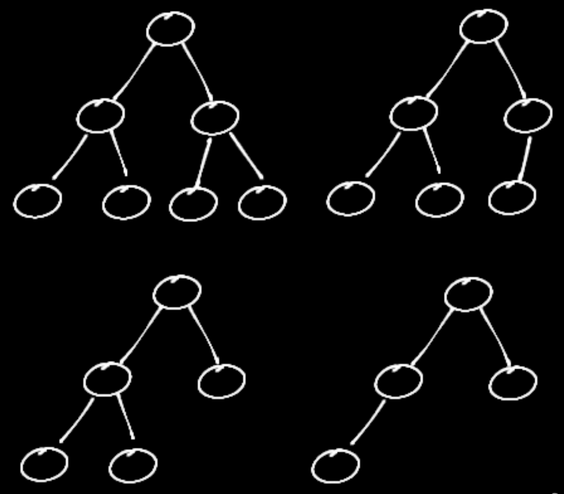


max height of the left and right side differ by more than one

Binary Search tree



Balanced Tree

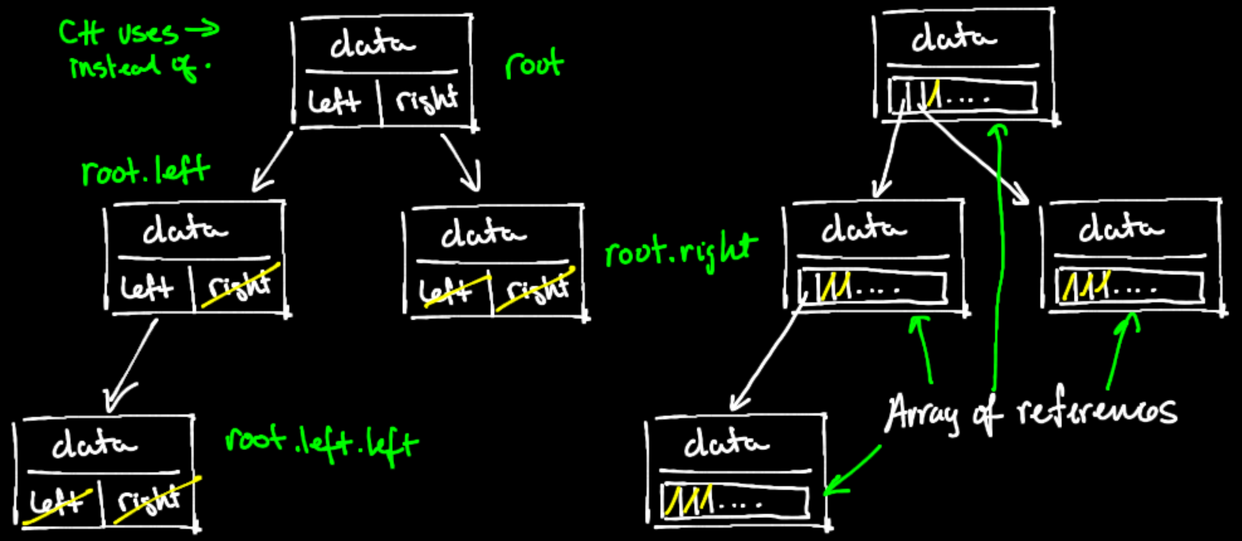


Tree of minimal height - max height of left and right subtrees differs by at most one

→

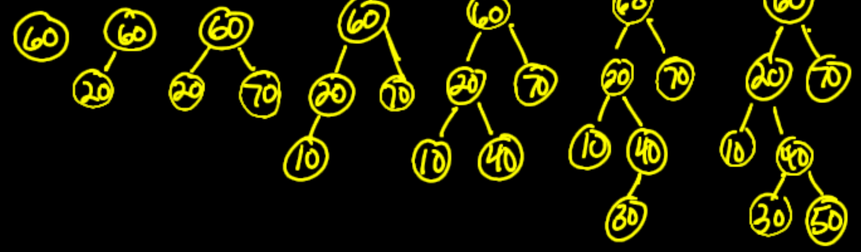
④ Reference-based

C++ uses → instead of .

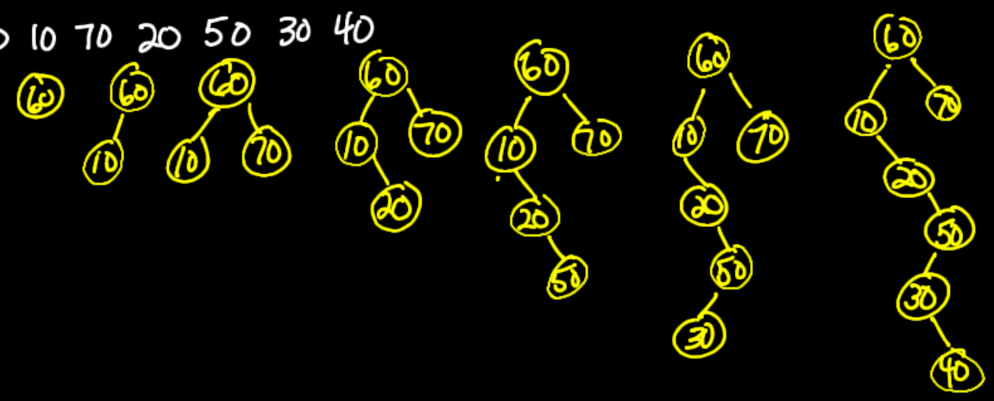


⑤ BST Insert

60 20 70 10 40 30 50



60 10 70 20 50 30 40



⑥ BST Insert

10 20 30 40 50 60 70



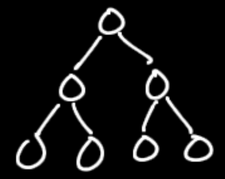
BST Performance

Worst case



O(n)

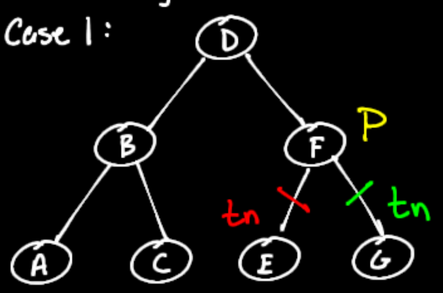
best case



$O(\log_2 n)$

⑦ Deleting Nodes

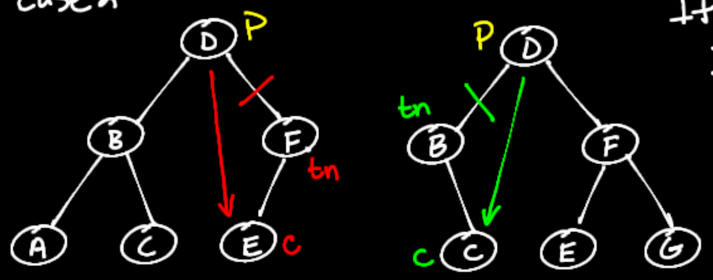
Case 1:



Let tn be the node to delete
Let p be the parent
Let c be the child
If (tn is a leaf)
  If (tn is the left of P)
    p.left = null
  else
    p.right = null

eg. Delete E
eg. Delete G

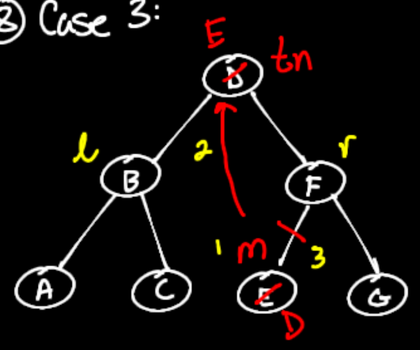
Case 2:

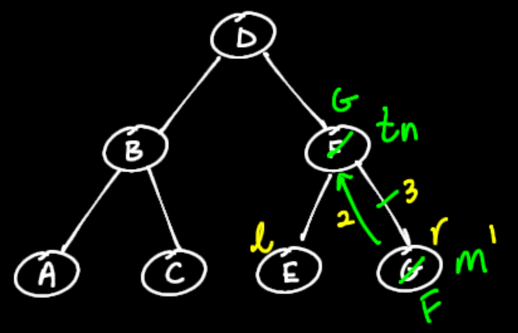


If (tn has one child)
  If (tn is the left of P)
    P.left = C
  else
    P.right = C

eg. Delete F
eg. Delete B

⑧ Case 3:





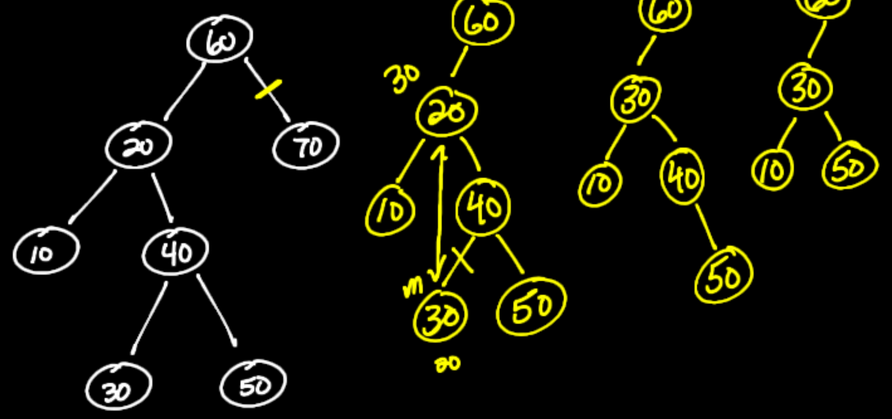
If (tn has two children)
  find M, the inorder successor of tn
  Copy M into tn
  remove M from the tree

Inorder successor
- Left most node of the right subtree of tn

Removing M alway results in Case 1 or 2

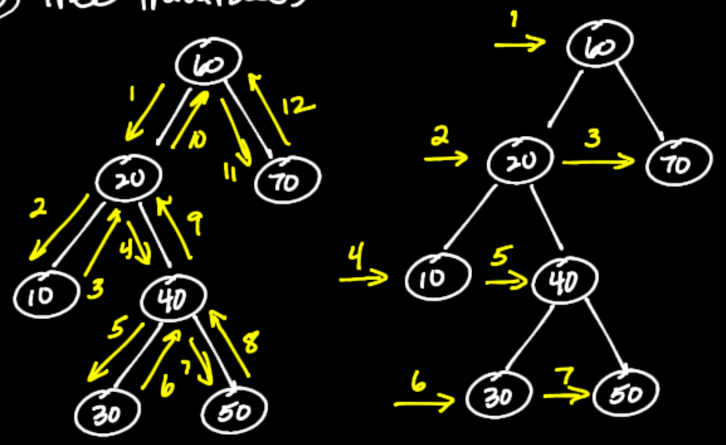
e.g. Delete D
eg. Delete F

(9) Delete 70, 20, 40 in that order



→

(10) Tree Traversals



Depth First – "to the leaves"
- Pre order - visit, left, right
- in order - left, visit, right
- post order - left, right, visit

Breadth First - "levels"
- Left to right

pre order: 60 20 10 40 30 50 70

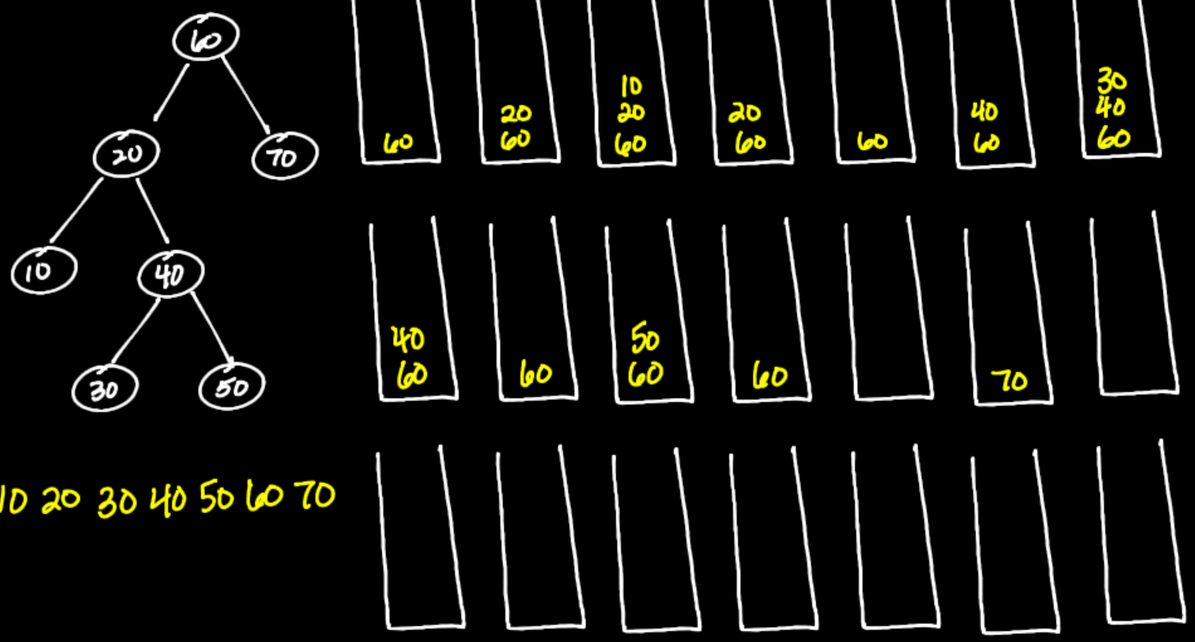
in order: 10 20 30 40 50 60 70

post order: 10 30 50 40 20 70 60

breadth first: 60 20 70 10 40 30 50

But how can we do this in code?

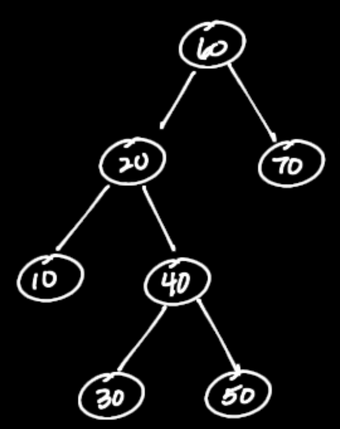
(11) Inorder w/ a stack



10 20 30 40 50 60 70

→

(12) Breadth First w/ a Queue



f ——— b

~~60~~ ~~20~~ ~~70~~ ~~10~~ ~~40~~ ~~30~~ ~~50~~

60 20 70 10 40 30 50